

Aim: To write an Embedded C program for interfacing Gas Sensor MQ-2 using Arduino Uno and Proteus

objective:

1. Simulate the **MQ-2 gas sensor** and **Arduino** circuit inside the **Proteus** software environment.
2. Interface the analog gas sensor with the **Arduino analog input** pin to read voltage changes.
3. Program the Arduino to **detect threshold gas levels** and trigger an output indicator like an LED or buzzer.

program:

```
#define GreenLed 13

#define Sensor A0


void setup(){
pinMode(13,OUTPUT);
pinMode(12,OUTPUT);
pinMode(A0,INPUT);
Serial.begin(9600);
}

void loop() {
int value = analogRead(A0);
Serial.print("Analogic Value coming from the sensor : ");
Serial.println(value);
delay(100);
if(value>600){
digitalWrite(13,HIGH);
}
else{
digitalWrite(13,LOW);
}
delay(20);
```

```
}
```

Code for MQ6 sensor:

```
int mq6Pin = A0; // MQ-5 analog pin connected to A0
```

```
void setup() {
```

```
Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
int sensorValue = analogRead(mq5Pin);
```

```
Serial.print("MQ5 Sensor Value: ");
```

```
Serial.println(sensorValue);
```

```
delay(1000); // 1-second delay
```

```
}
```

Output:

